



**Faculty of Engineering & Technology**

**Department of Electrical & Computer Engineering.**

**COMP433, SOFTWARE ENGINEERING**

**Phase Two:**

**Requirement Engineering- Elicitation & Discovery**

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**Instructor: Dr. Mohammed Husseinali**

**Section: 1.**

**Date: May– 2026**

## Task 2.1: USER REQUIREMENTS

Table 1: Task 2.1 Leadership and Contributions

| Role        | Name             | Contribution                  |
|-------------|------------------|-------------------------------|
| Lead        | Roa Makhtoob     | Writing and coordination      |
| Contributor | Abeer Hussein    | Reviewing                     |
| Contributor | Shaymaa Dar Taha | Discussing & coordination     |
| Contributor | Marwa Fageeh     | Feasibility feedback & review |

### UR1: Fresh flower browsing

The system shall allow customers to browse fresh flowers, including individual Bouquet, seasonal selections, and imported flowers.

### UR2: Order placement

The system shall allow customers to place orders for fresh flowers and pre-designed floral arrangement packages.

### UR3: Custom arrangement design

The system shall allow customers to customize floral arrangements based on occasion, preferences, and budget, with the option to request on-site setup or installation.

### UR4: Delivery services

The system shall provide local delivery services within the city, including a same-day delivery option subject to an additional fee .

### UR5: Subscription services

The system shall allow customers to subscribe to a recurring flower delivery service with flexible weekly, monthly, or yearly plans and customizable schedules.

### UR6: Scheduled delivery

The system shall allow customers to schedule delivery for a future date and time.

### UR7: Payment methods

The system shall support multiple payment methods, including cash (in-store), credit cards, and online payments.

### UR8: Consultation booking

The system shall allow customers to book consultation sessions (in-store or virtual) for floral design assistance.

### UR9: Workshop registration

The system shall allow customers to register for floral workshops (online or in-store) at beginner, intermediate, or advanced levels.

### UR10: Customer account management

The system shall allow customers to create and manage a personal account to view order history, track deliveries, manage subscriptions, and save personal preferences, delivery addresses and submit reviews and rating.

## Task 2.2: SYSTEM REQUIREMENTS

Table2: Task 2.2 Leadership and Contributions

| Role        | Name             | Contribution              |
|-------------|------------------|---------------------------|
| Lead        | Abeer Hussein    | Writing and coordination  |
| Contributor | Roa Makhtoob     | Requirement alignment     |
| Contributor | Marwa Fageeh     | Technical vaildation      |
| Contributor | Shaymaa Dar Taha | Coordination & finalising |

### UR1: Fresh flower browsing

**SR1.1:** The system shall display flowers organized into categories such as bouquets, seasonal flowers, and imported flowers.

**SR1.2:** The system shall display flower details including name, price, image, description, and availability, and allow customers to view full details of a selected flower.

**SR1.3:** The system shall allow customers to search for flowers by name or keywords.

**SR1.4:** The system shall allow customers to filter flowers based on category, price range, and occasion.

**SR1.5:** The system shall indicate whether a flower is available or out of stock.

**SR1.6:** The system shall allow staff to assist customers in browsing and viewing flower details in-store

## **UR2: Order placement**

**SR2.1:** The system shall allow customers to add flowers or arrangements to a shopping cart (online orders).

**SR2.2:** The system shall allow staff to create orders for customers in-store.

**SR2.3:** The system shall allow customers or staff to update or remove items from the order.

**SR2.4:** The system shall display the total price of the order.

**SR2.5:** The system shall allow customers or staff to confirm and submit an order.

**SR2.6:** The system shall generate a unique order ID for each confirmed order.

**SR2.7:** The system shall store order details including items, quantities, and total cost to prevent data loss during system failure.

**SR2.8:** The system shall ensure that an order contains at least one item before confirmation.

**SR2.9:** The system shall confirm and process orders within 3 seconds under normal operating conditions.

### **UR3: Custom arrangement design**

**SR3.1:** The system shall allow customers to enter and select details for a custom arrangement, including occasion, preferences, colors, budget, and desired flower types or styles.

**SR3.2:** The system shall allow customers to request on-site setup or installation.

**SR3.3:** The system shall store custom arrangement requests for processing by the florist

**SR3.4:** The system shall allow staff to review and manage custom arrangement requests.

**SR3.5:** The system shall allow staff to update the status of custom orders (pending, confirmed, completed).

**SR3.6:** The system shall allow staff to provide confirmation or feedback for custom arrangement requests.

### **UR4: Delivery services**

**SR4.1:** The system shall allow customers or staff to enter delivery details including address, contact information, and delivery instructions.

**SR4.2:** The system shall validate that the delivery address is within the supported delivery area.

**SR4.3:** The system shall allow customers or staff to select delivery type (standard or same day).

**SR4.4:** The system shall apply an additional fee for same-day delivery.

**SR4.5:** The system shall store delivery details with the order.

**SR4.6:** The system shall allow staff to manage and update delivery status (scheduled, out for delivery, delivered).

#### **UR5: Subscription services**

**SR5.1:** The system shall allow customers to select a subscription delivery frequency (weekly, monthly, or yearly).

**SR5.2:** The system shall allow customers or staff to define the subscription duration, including start date and end date.

**SR5.3:** The system shall allow customers or staff to customize subscription delivery schedules, including preferred delivery dates, times, addresses, and flower preferences.

**SR5.4:** The system shall allow customers to choose a subscription tier (Basic or Premium).

**SR5.5:** The system shall assign and automatically apply subscription benefits based on the selected tier, including discounts, delivery advantages, priority processing, and access to seasonal or exclusive flowers.

**SR5.6:** The system shall allow customers or staff to pause, modify, renew, or cancel a subscription.

**SR5.7:** The system shall store subscription details including delivery frequency, duration, schedule, preferences, and tier.

**SR5.8:** The system shall allow staff to view and manage customer subscriptions.

**SR5.9:** The system shall notify customers about upcoming subscription deliveries and subscription expiration dates

#### **UR6 Scheduled delivery**

**SR6.1:** The system shall allow customers or staff to select a future delivery date.

**SR6.2:** The system shall validate that the selected delivery date is valid and available.

**SR6.3:** The system shall store scheduled delivery details with the order.

**SR6.4:** The system shall allow staff to view and manage scheduled deliveries.

#### **UR7: Payment methods**

**SR7.1:** The system shall allow customers or staff to select a payment method for an order.

**SR7.2:** The system shall support cash payment for in-store orders.

**SR7.3:** The system shall support credit card payments.

**SR7.4:** The system shall support online payment methods.

**SR7.5:** The system shall confirm payment before completing an order.

**SR7.6:** The system shall store payment details associated with each order.

**SR7.7:** The system shall store payment information securely in encrypted form associated with each order.

#### **UR8: Consultation booking**

**SR8.1:** The system shall display available consultation time slots.

**SR8.2:** The system shall allow customers or staff to select a consultation type (in-store or virtual).

**SR8.3:** The system shall allow customers or staff to select a consultation date and time.

**SR8.4:** The system shall validate that the selected time slot is available.

**SR8.5:** The system shall store consultation booking details including date, time, and type.

**SR8.6:** The system shall allow staff to view and manage consultation bookings.

### **UR9: Workshop registration**

**SR9.1:** The system shall display available workshops including level, date, time, and price.

**SR9.2:** The system shall allow customers or staff to select a workshop type (online or in-store).

**SR9.3:** The system shall allow customers or staff to register for a selected workshop.

**SR9.4:** The system shall validate that workshop capacity is not exceeded before registration.

**SR9.5:** The system shall store workshop registration details including participant information and selected workshop.

**SR9.6:** The system shall allow staff to view and manage workshop registrations.

### **UR10: Customer account management**

**SR10.1:** The system shall allow customers to create and log in to an account.

**SR10.2:** The system shall allow customers to view their order history.

**SR10.3:** The system shall allow customers to track the status of their deliveries.

**SR10.4:** The system shall allow customers to manage their subscription settings.

**SR10.5:** The system shall allow customers to save and update delivery addresses and personal preferences.

**SR10.6:** The system shall allow customers to submit ratings and reviews for completed orders.

**SR10.7:** The system shall store customer account information securely.

**SR10.8:** The system shall allow staff to view customer account details when needed.

**SR10.9:** The system shall support access through modern web browsers including Chrome, Firefox, Edge, and Safari.

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## Task 2.3: EFFORT&TIME ESTIMATION

Table2: Task 2.3 Leadership and Contributions

| Role        | Name             | Contribution                    |
|-------------|------------------|---------------------------------|
| Lead        | Shaymaa Dar Taha | Writing and coordination        |
| Contributor | Abeer Hussein    | System complexity estimation    |
| Contributor | Marwa Faqeeh     | Development effort estimation   |
| Contributor | Roa Makhtoob     | Scope clarification & reviewing |

Acronyms: **pw** = person week, **pm** = person month, **w** = week, **m** = month.

Assumptions:

1 week = 5 days

1 working day = 8 hours

Average developer salary =\$250 per week

Schedule time includes an additional 30% buffer for testing, meetings, fixes, and unexpected

| UR                                 | Estimated No of Developers | Estimated Effort (Average/developers) | Total Effort (for one Developer) |
|------------------------------------|----------------------------|---------------------------------------|----------------------------------|
| UR1 – Fresh flower browsing        | 2                          | 2 pw                                  | $2*2= 4$ pw                      |
| UR2 – Order placement              | 3                          | 3 pw                                  | $3*3= 9$ pw                      |
| UR3 – Custom arrangement design    | 2                          | 3 pw                                  | $2*3= 6$ pw                      |
| UR4 – Delivery services            | 2                          | 2 pw                                  | $2*2= 4$ pw                      |
| UR5 – Subscription services        | 3                          | 4 pw                                  | $3*4= 12$ pw                     |
| UR6 – Scheduled delivery           | 2                          | 2 pw                                  | $2*2= 4$ pw                      |
| UR7 – Payment methods              | 3                          | 3 pw                                  | $3*3= 9$ pw                      |
| UR8 – Consultation booking         | 2                          | 2 pw                                  | $2*2= 4$ pw                      |
| UR9 – Workshop registration        | 2                          | 2 pw                                  | $2*2= 4$ pw                      |
| UR10 – Customer account management | 3                          | 4 pw                                  | $3*4= 12$ pw                     |
| <b>Total: 10</b>                   | <b>24</b>                  | <b>27 pw</b>                          | <b>68 pw</b>                     |

### Schedule Time Estimation:

Minimum Completion Time Using average effort:  $27 * 1.30 = 35$  weeks

Maximum Completion Time Using total effort:  $68 * 1.30 = 88$  weeks

### Cost Estimation:

Estimated project cost:  $250 * 88 = \$22000$

Profit Margin Estimation:

Minimum Cost with 10% Profit =  $22000 * 1.10 = \$24200$

Maximum Cost with 30% Profit =  $22000 * 1.30 = \$28600$

Table3: Task 2.2 Leadership and Contributions

| Role | Name         | Contribution                                   |
|------|--------------|------------------------------------------------|
| Lead | Marwa Faqeeh | Document Writing & Customer Meeting Management |

### Minutes of meetings:

- **Date:** 2/5/2026
- **Time:** 9:00 PM
- **Attendees:**

#### Developer Group

|                  |         |
|------------------|---------|
| Marwa Faqeeh     | 1220039 |
| Roa Makhtoob     | 1221636 |
| Shaymaa Dar Taha | 1222440 |
| Abeer Hussein    | 1220425 |

#### Customer Group

|                 |         |
|-----------------|---------|
| Abdallah Kokash | 1220116 |
| Bilal Ashhab    | 1220307 |
| Hanna Kaibni    | 1220214 |
| Jamil Habash    | 1220194 |

- **Topic of Discussion:** Discussing user and system requirements for Phase 2.



Figure 1: Online Meeting with Customer Group

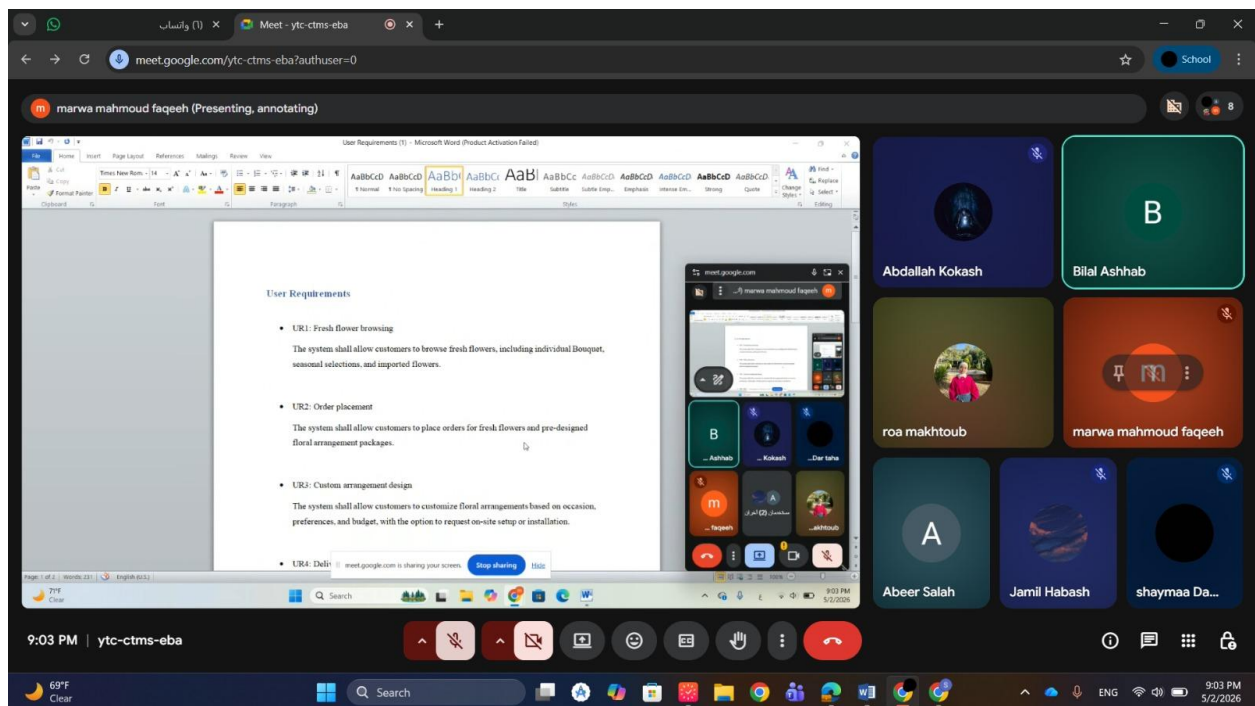


Figure 2: Requirements Discussion Meeting

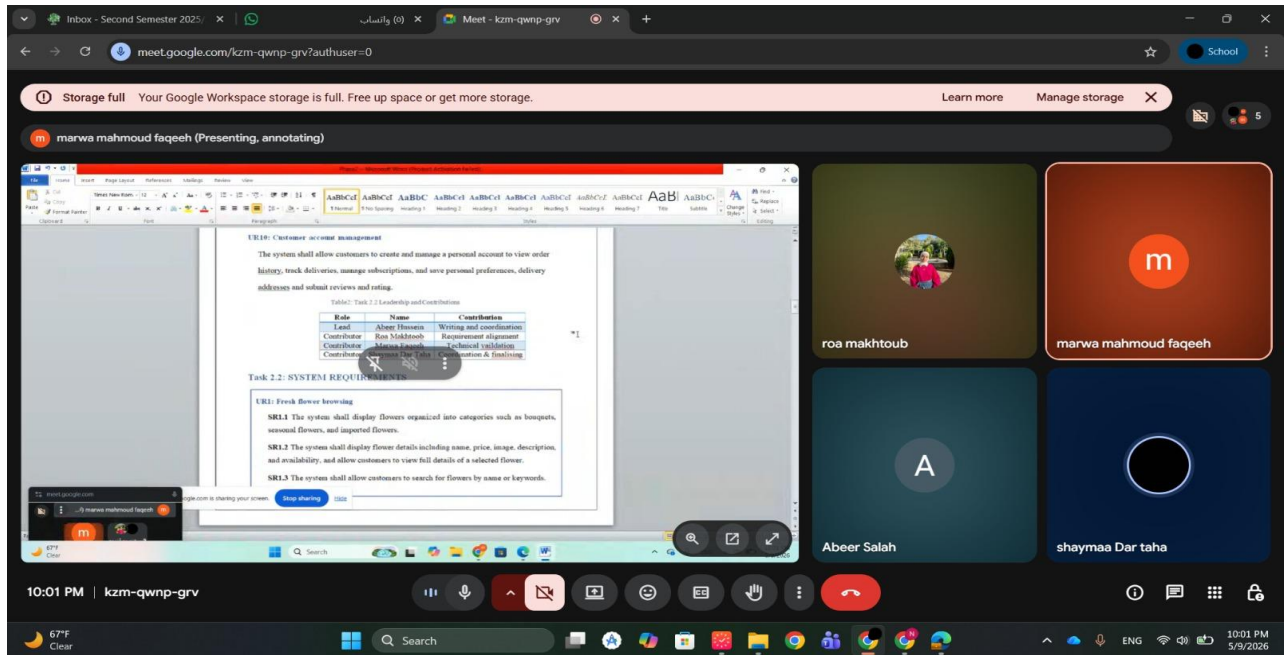


Figure 3:meeting to finish the task with our group